

26 > Optimise devices for better performance

■ Select the Best Channel:

- Use Wi-Fi analysis tools to find the least congested channel.

How to reduce network load

Managing your network devices can prevent congestion and improve performance.

Prioritising devices and bandwidth

■ Quality of Service (QoS):

- Configure QoS settings in your router to prioritize bandwidth for critical devices or applications.

■ Limit Background Activity:

- Pause or schedule updates and backups during non-peak times.

Using wired connections for key devices

■ Ethernet Connection:

- Provides a stable and faster connection for PCs, consoles, or smart TVs.

■ Powerline Adapters:

- Use existing electrical wiring to extend wired connections.

Constantly monitor network health

Regular monitoring can help identify and resolve issues promptly.

How to monitor data usage and speed

■ Speed Tests:

- Ookla Speedtest, Fast.com to measure internet speeds.

■ Network Monitoring Apps:

- GlassWire, NetWorx for tracking data usage.

5) Backing up and protecting device data

Safeguarding your data against loss, theft, or damage is essential. Implementing regular backups and security measures ensures your information remains secure and accessible.

Cloud vs. Physical backup

Choosing the right backup method depends on your needs. Both cloud and physical backups have their advantages and disadvantages.

Cloud storage vs. External hard drives: Pros vs cons

Cloud storage

■ Pros:

- **Accessibility:** Access files from any device with internet connectivity.
- **Automatic Backups:** Set and forget; backups occur in the background.
- **Offsite Storage:** Protects data from physical disasters.

■ Cons:

- **Recurring Costs:** Monthly or annual fees for storage space.
- **Security Risks:** Potential vulnerability to hacking.
- **Dependence on Internet:** Requires a stable connection.

External hard drives

■ Pros:

- **One-Time Cost:** No ongoing fees.
- **Control:** Data remains physically with you.
- **Speed:** Faster data transfer rates, especially with USB 3.0 or Thunderbolt.

■ Cons:

- **Physical Damage or Loss:** Susceptible to hardware failure.
- **Manual Effort:** Requires regular connection and backup initiation.
- **Limited Accessibility:** Data isn't accessible remotely.

Recommendations for backup frequency

■ Regular Backups:

- **Daily:** For critical data or frequent changes.
- **Weekly:** For general use.

■ Automate When Possible:

- Use backup software to schedule regular backups.

■ Multiple Backups:

- Combine cloud and physical backups for redundancy.

Protecting your data from loss and theft

Implementing security measures can prevent unauthorised access and protect your data.

Encryption options for personal devices

■ Encrypt Hard Drives:

● Windows:

- **BitLocker:** Available on Pro and Enterprise editions.
- **Enable:** Control Panel > System and Security > BitLocker Drive Encryption.

● macOS:

- **FileVault:** System Preferences > Security & Privacy > FileVault.

■ Use Encrypted Storage Devices:

- Purchase external drives with built-in encryption.
- Using biometric and password security

■ Strong Passwords:

- Use complex passwords with a mix of characters.
- Avoid common words or easily guessable information.

■ Biometric Authentication:

- **Fingerprint Scanners:** Secure and convenient.
- **Facial Recognition:** Available on newer devices.

■ Two-Factor Authentication (2FA):

- Add an extra layer of security to your accounts and devices.

■ Lock Screen Settings:

- Set devices to lock after a short period of inactivity.
- Use secure lock methods (PIN, password, biometrics). 